

DeRexi: Policy Pilot

VULNERABILITY ASSESSMENT REPORT

Network discovery • service enumeration • vulnerability analysis

Assessment platform Kali Linux 2023 | Tool Nmap 7.94

DISCOVERY SCOPE

10.11.2.0/24

ACTIVE HOSTS

10

ASSESSED ASSET

10.11.2.144

SCENARIO NOTE The IP address and scan results originate from an authorized classroom lab. For this report, the assessed host is treated as a fictional DeRexi internal application server so the methodology can be applied within the project scope.

Executive Summary

A vulnerability assessment was conducted from a Kali Linux security workstation using Nmap 7.94. The assessment combined network asset discovery, full TCP port enumeration, service identification, and Nmap Scripting Engine vulnerability checks. Within the fictional DeRexi scenario, the selected host at 10.11.2.144 represents a critical internal application server.

The discovery scan identified 10 active hosts within the tested 10.11.2.0/24 range. Detailed assessment of the selected host examined TCP ports 1-65535 and found one accessible service: SSH on TCP/22. Nmap's vulnerability scripts did not establish a confirmed target-specific vulnerability. The principal security observation is therefore the exposure of a remote administrative service that warrants continued hardening, access restriction, patch management, and monitoring.

Key Findings

Finding	Evidence	Classification	Severity
Asset discovery	10 active hosts identified in 10.11.2.0/24	Discovery result	Informational
SSH exposure	TCP/22 open on 10.11.2.144	Security exposure / review	Low / Informational
Vulnerability scripts	No confirmed target vulnerability shown	Assessment result	Informational
CVE-2011-1002 check	Pre-scan output explicitly reported hosts not vulnerable	Not a finding	N/A

1. Assessment Methodology

The assessment followed a staged process: identify reachable assets, select an authorized target, enumerate the target's TCP services, and then run vulnerability-oriented NSE checks. This approach reduces guesswork and ties later vulnerability analysis to observed network exposure.

1.1 Asset Discovery

Command: `nmap -sn 10.11.2.0/24`

The `-sn` option performs host discovery without a port scan. Nmap assessed the specified /24 range and reported 10 active hosts. This phase established which systems were reachable before deeper assessment.

1.2 Service and Port Enumeration

Command: `nmap -sV -sC -p1-65535 10.11.2.144`

The `-sV` option probes open ports for service/version information; `-sC` runs Nmap's default NSE scripts; and `-p1-65535` directs Nmap to examine the full numbered TCP port range. The scan reported 65,534 closed TCP ports and one open port: TCP/22 running SSH.

1.3 Vulnerability Script Scan

Command: `nmap -sC --script=vuln 10.11.2.144`

The NSE vulnerability-category scripts were used to check applicable services for known vulnerability conditions. The visible output did not establish a target-specific vulnerability affecting SSH. A pre-scan Avahi check referenced CVE-2011-1002 but explicitly stated that the tested hosts were not vulnerable; it is therefore not reported as a vulnerability of the assessed DeRexi asset.

2. Critical Asset Identification

For the DeRexi scenario, 10.11.2.144 is classified as the DeRexi Internal Application Server. It is treated as a critical asset because DeRexi: Policy Pilot is designed to support centralized organizational policy access, employee authentication, policy Q&A, policy-owner routing, and incident-assistance workflows.

Compromise of this server could affect the confidentiality of organizational information, the integrity of policy or application content, and the availability of employee-facing services. These potential business impacts justify additional scrutiny of any remotely accessible administrative service.

3. Findings and Classification

VA-01 | Exposed SSH Administrative Service

Classification: Security exposure / configuration review **Severity:** Low / Informational pending further validation

TCP port 22 was open and identified as SSH (Secure Shell). SSH is a legitimate encrypted remote-administration protocol; its presence alone is not a vulnerability. Because it can provide administrative access, however, unnecessary exposure or weak configuration could increase the attack surface of the server.

The scan alone does not establish whether password authentication, key-based authentication, MFA, firewall restrictions, administrator allowlists, or current patching are in place. Those controls require separate validation and should not be assumed absent.

VA-02 | No Confirmed Vulnerability Detected

Classification: No confirmed vulnerability **Severity:** Informational

Nmap's vulnerability scripts did not identify a confirmed vulnerability affecting the assessed SSH service in the captured results. The Avahi-related CVE reference in the pre-scan output was explicitly marked not vulnerable and is not treated as a positive finding.

4. Potential Security Implications

Security Objective	Potential Impact if Administrative Access Were Compromised
Confidentiality	Unauthorized access to internal policy information or other protected application data.
Integrity	Unauthorized modification of application configuration, policy content, or supporting system components.
Availability	Disruption of DeRexi services, potentially preventing employees from accessing policy or incident-assistance functionality.

5. Recommended Security Controls

Restrict administrative access Limit SSH to authorized administrators and trusted management networks or approved source addresses.

Use strong authentication Prefer key-based authentication and additional administrative controls where supported; avoid weak or shared credentials.

Maintain patch currency Keep the SSH implementation and underlying operating system current with security updates.

Monitor authentication activity Review SSH authentication logs for repeated failures, unusual sources, and other indicators of unauthorized access attempts.

Reduce attack surface If SSH is not operationally required on the server, disable the service or block network access to TCP/22.

6. Assessment Conclusion

The assessment successfully demonstrated network discovery, critical-asset selection, full TCP port enumeration, service identification, and vulnerability-oriented scanning using Nmap. Ten active hosts were identified in the selected discovery range, and the designated DeRexi application server exposed one TCP service: SSH on port 22.

No confirmed target-specific vulnerability was established by the captured vulnerability scan. The assessment therefore classifies SSH as a security exposure requiring configuration review rather than as a confirmed vulnerability. DeRexi should maintain strict administrative-access controls, current patching, and monitoring around remote-management services.

Appendix A | Scan Evidence

Selected screenshots from the authorized Kali Linux classroom assessment are included below. (Duplicate captures were intentionally omitted.)

```
(kali@kali)-[~]
$ nmap -sn 10.11.2.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-01 13:23 PDT
Nmap scan report for 10.11.2.25
Host is up (0.0012s latency).
Nmap scan report for 10.11.2.74
Host is up (0.0094s latency).
Nmap scan report for 10.11.2.97
Host is up (0.00037s latency).
Nmap scan report for 10.11.2.144
Host is up (0.025s latency).
Nmap scan report for 10.11.2.155
Host is up (0.0077s latency).
Nmap scan report for 10.11.2.159
Host is up (0.0088s latency).
Nmap scan report for 10.11.2.177
Host is up (0.011s latency).
Nmap scan report for 10.11.2.193
Host is up (0.010s latency).
Nmap scan report for 10.11.2.205
Host is up (0.11s latency).
Nmap scan report for 10.11.2.226
Host is up (0.011s latency).
Nmap done: 256 IP addresses (10 hosts up) scanned in 8.56 seconds
```

Figure A1. Nmap host-discovery scan of 10.11.2.0/24 reporting 10 active hosts.

```
(kali@kali)-[~]
$ nmap -sV -sC -p1-65535 10.11.2.144
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-01 14:16 PDT
Nmap scan report for 10.11.2.144
Host is up (0.0097s latency).
Not shown: 65534 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      (protocol 2.0)
|_ fingerprint-strings:
|_  NULL:
|_  SSH-2.0-X17.1.2
|_  I service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/sub
mit.cgi?new-service:
SF-Port22-TCP:V=7.94%1=7ND-9/1%Time=6A9740C3%P=aarch64-unknown-linux-gnu%r
SF:(NULL,11,"SSH-2\0-X17\1\2\r\n");
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.15 seconds
```

Figure A2. Full TCP service scan of 10.11.2.144 showing TCP/22 open for SSH and 65,534 closed TCP ports.

```
(kali@kali)-[~]
$ nmap -sc --vulnscan 10.11.2.144
nmap: option '-sc' is ambiguous; possibilities: '-scanflags' '-scan-delay' '-script' '-script-trace' '-script-updatedb' '-script-args' '-script-args-fi
le' '-script-help' '-script-timeout'
See the output of nmap -h for a summary of options.

(kali@kali)-[~]
$ nmap -sC -script=vuln 10.11.2.144
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-01 14:23 PDT
Failed to resolve "script-vuln".
Nmap scan report for 10.11.2.144
Host is up (0.0064s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.95 seconds

(kali@kali)-[~]
$ nmap -sC -script=vuln 10.11.2.144
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-01 14:23 PDT
Pre-scan script results:
| broadcast-avahi-dos:
|_  Discovered hosts:
|_  224.0.0.251
|_  After NULL UDP avahi packet DoS (CVE-2011-1002).
|_  Hosts are all up (not vulnerable).
Nmap scan report for 10.11.2.144
Host is up (0.0076s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 36.4 seconds
```

Figure A3. Corrected Nmap vulnerability-script scan and resulting SSH service output.